

ROBOTICS (24/07/2019)

Matteo Matteucci

The exam will be graded IFF the following recommendations have been taken into account:

- Write clearly so that the teacher can easily understand your answers
- Write your name, surname, and student id on each sheet you deliver for evaluation
- For each exercise/question report clearly the number and sub-number (if present)
- You are not allowed to use any programmable device (e.g., smartphone, calculator, etc.)
- You can use pen or pencil, paper will be provided, you cannot use notes or books

Exercise 1 – Sensors and Actuators [1+1+2+2 points]

Mobile robots use sensors of different kinds:

- What is difference between proprioceptive and exteroceptive sensors?
- Provide one example of proprioceptive sensor and one exteroceptive sensor
- How does an ultrasound sensor (sonar) work? What are its pros and cons?
- How does a laser range finder (lidar) work? What are its pros and cons?

Exercise 2 – Robot Odometry [2+1+3 points]

The figure shows an electric vehicle with a quite common kinematics. With reference of this kind of vehicle answer the following questions:

- Compute the direct kinematics of the vehicle in the picture and discuss the relevant parameters of such formulas.
- What is inverse kinematics? Is it easy or not to compute a possible inverse kinematics for this kind of vehicle? Why?
- Let's measure v_f and ϕ the front wheel speed and the steering angle respectively, and assume the distance from the odometric center and the front wheel $L = \sqrt{3} m$. Assume the robot starts from position $[0, 0, 0]$ on the plane, where it will be after performing each of the following actions one after the other?
 - For 0.2 s keep $v_f = 2 m/s$, and $\phi = 0$
 - For π s keep $v_f = 1.5 m/s$, and $\phi = 30^\circ$
 - For π s keep $v_f = 3 m/s$, and $\phi = -30^\circ$



	30°	60°
sin	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$
cos	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$
tan	$\frac{1}{\sqrt{3}}$	$\sqrt{3}$

Exercise 3 – Robot Local Path Planning [2+1+2]

Let's consider the problem of obstacle avoidance, a.k.a., local path planning, and the algorithms to realize it.

- Provide a detailed description of the Vector Field Histogram (VFH) algorithm
- Describe the limits of VFH and how these are faced in VFH+
- Describe the algorithm DWA in its trajectory rollout variation. How this improves with respect to VFH and VFH+ algorithms?

Exercise 4 – Robot Operating System [2 + 2 points]

- What is the role of the ROS Master in a ROS architecture? Is the parameter server part of the ROS Master?
- What is the parameter server and what it is used for?

Exercise 5 – Simultaneous Localization and Mapping [1 + 1 + 1 + 2 + 1 points]

Let's consider the Bayes Filter in its recursive formulation:

- Provide the derivation of the standard recursive Bayes Filter
- What is the difference between online SLAM and full SLAM?
- How does the recursive Bayes Filter formula change from localization to online SLAM?
- Describe the FastSLAM algorithm for full SLAM
- What is the memory complexity of the full SLAM algorithm in case of a grid based map?